TECHNOLOGY STACK

Architecture & Stack | Project Design Phase-II

Date	2 November	**Team ID**	_____
Project Name	Electric Vehicle Performance and Insights	**Maximum Marks**	2 Marks

Define Your Technology Stack

The project uses a lightweight analytics architecture centred on a CSV source file and Tableau Desktop. The table below identifies the technologies actually used and explains their role in the solution.

S.No.	Component	Description	Technology	Justification / Purpose
1	Programming / Query Language	Language or calculation logic used to query, filter, transform, aggregate and analyse data.	Tableau calculated fields and built-in aggregation functions.	Used to calculate KPI measures and analytical metrics within the Tableau workbook. No separate Python or SQL implementation is claimed.
2	Database / Data Warehouse	Location and format in which structured project data is supplied.	CSV file: `ev_market_train.csv`.	The project dataset is provided as a flat file and connected directly to Tableau. No separate database or data warehouse is used.
3	Data Ingestion	Method used to load source data into the analytical environment.	Tableau Desktop CSV data connection.	Provides access to the EV market records and fields required for the worksheets and dashboard.
4	Data Transformation and Business Logic	Tools and logic used to prepare, aggregate and calculate analytical measures.	Tableau Desktop data handling, aggregations and calculated fields.	Supports measures such as Sales Contribution %, Average Sales per Vehicle and EV Customer Rating Status, along with KPI aggregations.
5	Data Visualization / Analytics	Tool used to create charts, KPIs, dashboards and analytical stories.	Tableau Desktop.	Used to create six worksheets, four KPI views, the interactive EV Market Performance Dashboard and the seven-point Tableau Story.
6	Front-End / User Interface	Interface through which users interact with dashboard content and filters.	Tableau dashboard and story interface.	Allows users to view KPIs and charts, apply Market Region and Model Year filters, and navigate the story.

High-level architecture

Data source	Analytics layer	Presentation layer
`ev_market_train.csv` EV market records	Tableau Desktop CSV connection, aggregations, calculated fields and worksheets	EV Market Performance Dashboard 4 KPI views + 6 charts Market Region / Model Year filters 7-point Tableau Story

Technology selection rationale

Tableau Desktop is suitable for this project because the required outcome is an interactive business-intelligence dashboard with KPI summaries, comparative charts, filters and a guided story. A CSV connection keeps the data source straightforward for an academic project. The current stack intentionally avoids naming tools or integrations that were not used.